Computational Statistical Physics
Exercise 3

Alexander Resare

March 2025

1 Description

This report builds on the previous task of simulating the 2D Ising model to gain insights on it's qualitative behavior near phase transitions. Particularly, the Wolff and Swedsen-Wang cluster algotihms will be considered. For different lattice configurations at and near the critical temperature T_C, the phenomena of critical slowing down shall be investigated in terms of how it affects the linear autocorrelation times for Metropolis compared to the cluster algorithms. To clarify, T_C refers again to Onsager's exact solution of the 2D Ising model:

$$\frac{k_B T_C}{J} = \frac{2}{\ln(1+\sqrt{2})} \approx 2.269 \tag{1}$$

2 Observables

For a temperature range $T \in [1.5, 3.0]$ with a step size 0.1, a 32×32 lattice is simulated via each algorithm and a selection of observables are plotted. This selection considers the absolute magnetization $|M|$ (normalized to the lattice size), the magnetization susceptibility χ and the Binder cumulant defined as:

$$U_L = 1 - \frac{\langle M^4 \rangle}{3\langle M^2 \rangle^2} \tag{2}$$

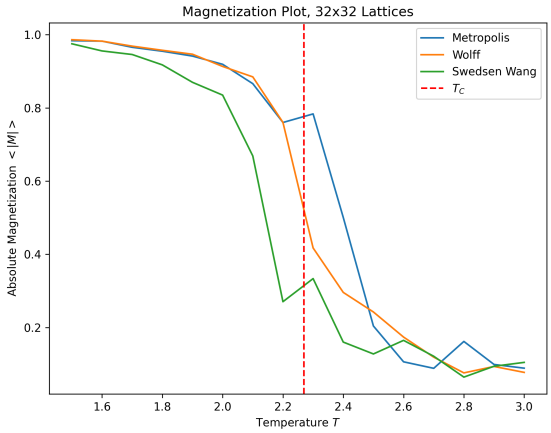

Figure 1: Magnetization in temperature range $T \in [1.5, 3]$

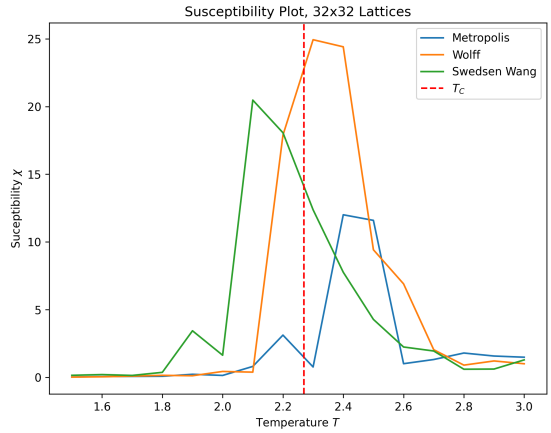

Figure 2: Susceptibility in temperature range $T \in [1.5, 3]$

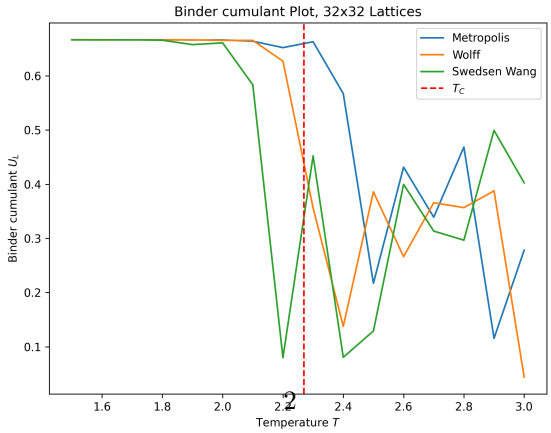

Figure 3: Binder cumulant in temperature range $T \in [1.5, 3]$

Comparing each plot, the phase transition for the Swedsen-Wang lattice seems occur before T_C, where as in the case of Metropolis the transition is slightly delayed. This may very well be a result of finite-size effects or the limited number of iterations. It is however possible to conclude that each algorithm adheres to predicted behavior and in every case before the phase transition, $U_L = 2/3$ as expected.

3 Linear Correlation Times for different Temperatures

3.1 Calculations and Plotting via Binning Method

This section will mainly consider the spin linear correlation time τ_σ, denoting the equilibration relaxation time for the lattice, given an initial configuration and fixed temperature. Later on, the Monte Carlo speed MC_{speed} shall also be defined and considered. The linear correlation time builds upon the autocorrelation, which is defined as the following:

$$\Phi_\sigma(\tau) = \frac{\langle \sigma(\tau_0)\sigma(\tau)\rangle - \langle \sigma(\tau_0)\rangle^2}{\langle \sigma^2(\tau_0)\rangle - \langle \sigma(\tau_0)\rangle^2} \tag{3}$$

From here, one can define the linear correlation time as:

$$\tau_\sigma = \int_0^\infty \Phi_\sigma(\tau)\mathrm{d}\tau \tag{4}$$

Equation (3) and (4) provides mainly a theoretical basis for interpreting τ_σ. As means of calculations, the Binning method shall be used. The details regarding this method will not be covered here although it shall be noted that the minimum number of bins used in each case was 32. In each plot, quantities Δ_l have been generated in a range $l \in [0, l_{max}]$ with integer step size. The linear autocorrelation time is then simply calculated as:

$$\tau_\sigma = \arg\max_l \frac{\Delta_l}{\Delta_0} \tag{5}$$

The temperature range considered here is $T \in \{T_C - 1, T_C - 0.5, \ldots, T_C + 1\}$. For each temperature, the peak of the curve is marked, as this is the Δ_l considered for the computation of τ_σ.

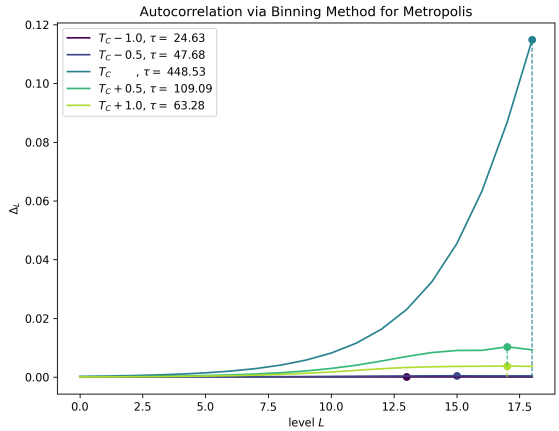

Figure 4: Linear autocorrelation times for 16×16 Metropolis simulated lattice.

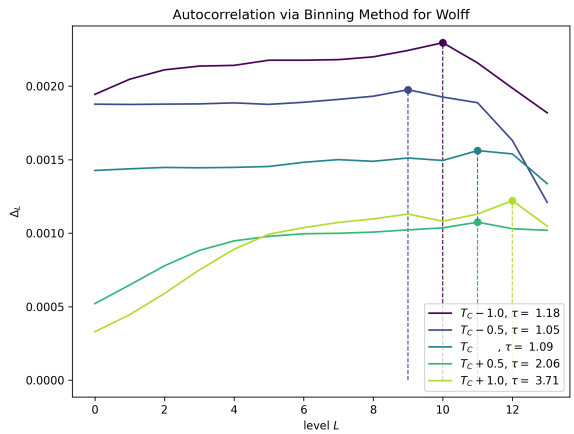

Figure 5: Linear autocorrelation times for 16×16 Wolff simulated lattice.

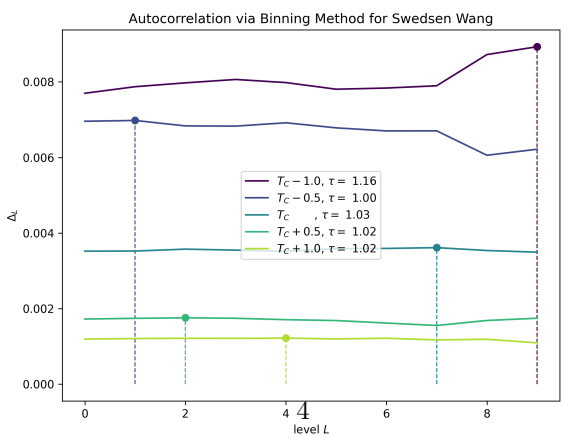

Figure 6: Linear autocorrelation times for 16×16 Swedsen-Wang simulated lattice.

In the case of Metropolis, the lattice seems to equilibrate around most temperatures for $l = 16$, where as in the case of Wolff around $l = 11$. For Swedsen-Wang the implied peak is more distributed across different levels. Since barely any clear peaks are visible, this could be a sign that the lattice simulated via Swedsen-Wang equilibrates much quicker, which could facilitate the necessity for a different method of computing the linear autocorrelation times. Finally, it is evident that Metropolis suffers from critical slowing down at T_C. Since no plateau is reached, it is easy to recognize that the linear autocorrelation time via this method actually diverges and the calculated value should thereby be discarded. The cluster methods on the other hand, seem to equilibrate well even at the critical temperature.

3.2 Monte Carlo Speed

The Monte Carlo speed is defined acording to the following formula:

$$MC_{speed} = \frac{\text{number of update steps}}{\text{runtime} \cdot \tau} \tag{6}$$

This quantity might vary depending on hardware and wether seconds or milliseconds are considered for the runtime. Therefore, the Monte Carlo speed here shall only be considered on a relative basis, to compare how the algorithms fares across different configurations. Worth noting is that the data listed in this report was simulated on a laptop with an Intel® Core™ i7-10510U CPU. To exclude unnecessary runtime from the Monte Carlo speed calculations, the linear autocorrelation times above are calculated anew, but with l_{max} reduced to where the peak of the corresponding curve is located. Then, given the minimum number of bins 32 as previously mentioned, the number of steps can be deduced:

$$\text{number of update steps} = 32 \cdot 2^{l_{max}} \tag{7}$$

Temperature		$T_C - 1.0$	$T_C - 0.5$	T_C	$T_C + 0.5$	$T_C + 1.0$
Metropolis	Runtime	0.14 s	0.56 s	3.66 s	1.89 s	1.97 s
	MC_{speed}	8.08e4	4.91e4	5.12e3	2.09e4	3.15e4
	l_{max}	13	15	18	17	17
Wolff	Runtime	10.44 s	4.26 s	8.62 s	1.03 s	0.82 s
	MC_{speed}	3.02e3	3.74e3	6.02e3	3.37e4	4.50e4
	l_{max}	10	9	11	11	12
Swedsen-Wang	Runtime	61.08 s	0.23 s	15.29 s	0.47 s	1.91 s
	MC_{speed}	2.31e2	3.34e2	2.90e2	2.48e2	2.35e2
	l_{max}	9	1	7	2	4

Table 1: Runtime listed in seconds and associated Monte Carlo speed for equilibration. The cells marked in yellow contains placeholder data. Since the Δ curve never reached a plateau, these values do not reflect valid measurements.

In terms of de-correlating the lattice, it is apparent that Swedsen-Wang is most efficient regarding the MC_{speed} metric. Wolff is also quite fast, more so

for temperatures below T_C. Finally, Metropolis is least efficient and fails to de-correlate the lattice at T_C. This may come as expected, due to the effect of critical slowing down.

4 Linear Correlation Times for different Lattice Sizes

4.1 Calculations and Plotting via Binning Method

This section aims to investigate how the linear autocorrelation times are affected by varying the lattice size across the different update methods. Specifically for this report, lattice sizes 10×10, 20×20 and 30×30 will be considered.

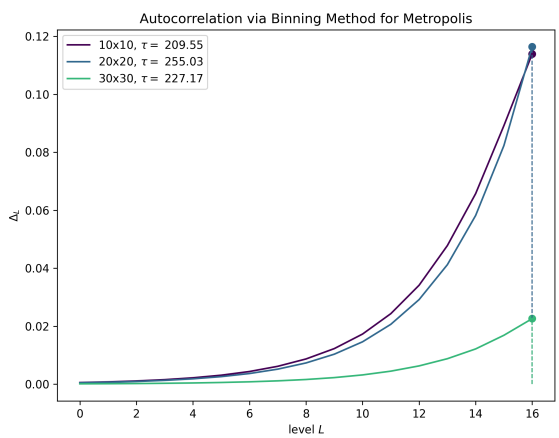

Figure 7: Linear autocorrelation times for Metropolis simulated lattices of varying sizes.

As expected, the autocorrelation times once again diverges at T_C for Metropolis-Hastings.

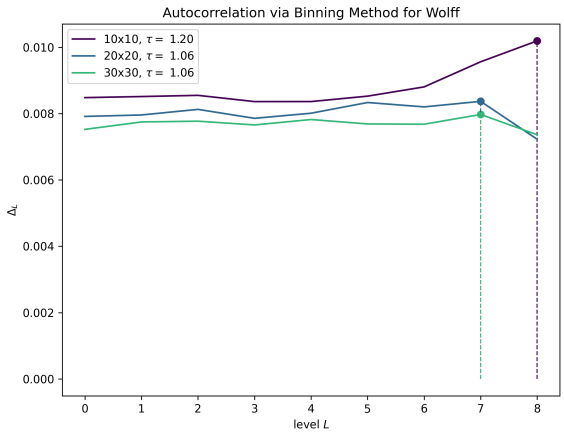

Figure 8: Linear autocorrelation times for Wolff simulated lattices of varying sizes.

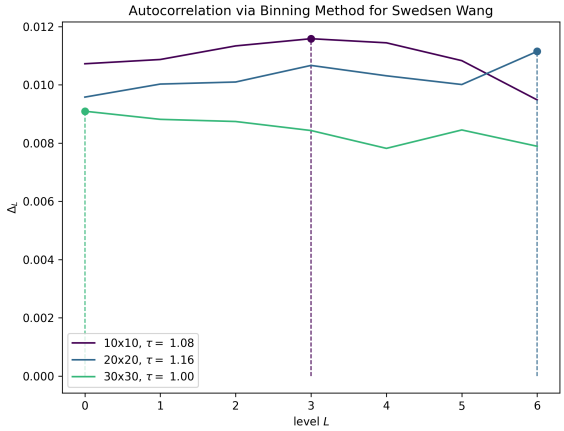

Figure 9: Linear autocorrelation times for Swedsen-Wang simulated lattices of varying sizes.

4.2 Monte Carlo Speed

Lattice size		10×10	20×20	30×30
Metropolis	Runtime	$0.95\ s$	$0.94\ s$	$0.93\ s$
	MC_{speed}	$1.05e4$	$8.79e3$	$9.97e3$
	l_{max}	16	16	16
Wolff	Runtime	$0.48\ s$	$0.77\ s$	$1.62\ s$
	MC_{speed}	$1.44e4$	$5.48e3$	$2.45e3$
	l_{max}	8	7	7
Swedsen-Wang	Runtime	$0.22\ s$	$16.93\ s$	$2.44\ s$
	MC_{speed}	$1.12e3$	$1.03e2$	25.1
	l_{max}	3	6	1

Table 2: Runtime listed in seconds and associated Monte Carlo speed for equilibration. Once again, it must be taken into consideration that the Metropolis measurements are not representative.

Interestingly enough, in the case of varying the lattice size at T_C, it seems the Wolff method performs better for larger sizes, where as for Swedsen-Wang it is the opposite behavior. No conclusions can be drawn from the yellow marked Metropolis data, which has just been listed for the sake of consistency.

5 Theoretical Description of Bond Probabilities

For the cluster algorithms in this report, the description of Ising model bond probabilities have been utilized. Here, the fundamental description of bond probabilities will be extended from the Ising model to also consider the Potts model. Given no external field, the Hamiltonians are given in each case:

$$\text{Ising:} \quad \mathcal{H} = -J \sum_{\langle i,j \rangle} \sigma_i \sigma_j \tag{8}$$

$$\text{Potts:} \quad \mathcal{H} = -J \sum_{\langle i,j \rangle} \delta_{\sigma_i \sigma_j} \tag{9}$$

Here, it is noted that:

$$\delta_{\sigma_i \sigma_j} = \begin{cases} 1, & \text{if } \sigma_i = \sigma_j \\ 0, & \text{otherwise} \end{cases} \tag{10}$$

The bond probabilities are related to the energy difference between two spins, given a spin flip. For both cases, assume two spins σ_1, σ_2 with possible states $\{-1, 1\}$. At first, the spins are aligned, leading to an initial energy $E_{1,2} = -J1 \cdot 1 = -J, -J\delta_{1,1} = -J$ respectively. In the case spin one is flipped such that $\sigma_{1'} = -1$, the new energy is $E_{1',2} = -J(-1) \cdot 1 = J, -J\delta_{-1,1} = 0$. This results in an energy difference $\Delta E_{1,2} = E_{1',2} - E_{1,2} = 2J$ (Ising) , J (Potts) respectively. This energy difference between spin flips serves as the basis for the bond probabilities, which leads to the final results:

$$\text{Ising:} \quad p = 1 - e^{-2J\beta} \tag{11}$$

$$\text{Potts:} \quad p = 1 - e^{-J\beta} \tag{12}$$